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Project Overview

LOAN MANAGEMENT SYSTEM – WEB APPLICATION FUNCTIONAL SPECIFICATION

1. Overview

The Loan Management System is a secure, role-based web platform that allows loan companies to manage customers, employees, and loan records centrally. All loan processing occurs outside the system; employees manually update loan amounts, interest, installments, and status. The platform provides role-based dashboards, real-time chat, admin monitoring, bilingual support (Arabic RTL & English), and strong access control.

2. User Roles & Permissions

Admin

Full system control. Can manage employees and customers, assign/reassign customers, view and edit all loans, monitor chats, access internal rooms, send announcements, manage languages, and view analytics.

Employee / Loan Officer

Access limited to assigned customers. Can create and update loans manually, change loan status, edit loan details at any stage, chat with customers, share files, access internal rooms, and receive notifications. Cannot access other employees' customers.

Customer

Read-only access to loan data. Can view loan details and status, chat with assigned employee, upload files, and receive notifications. Cannot edit loan information.

3. Authentication & Security

Secure login for all roles, role-based dashboards, session management, encrypted chat, secure file storage, and Arabic/English language support.

4. Customer Management

Admin can add, assign, and reassign customers and view profiles with chat history. Employees can view and manage only assigned customers. Customers can view their own profile and loan data.

5. Loan Management

All loan data is entered manually. No automatic calculations.
Loan creation includes amount, interest %, installments, totals, start date, notes, and optional payment schedule.
Statuses: Under Review, Approved, Active, Rejected, Closed.
Admin and Employee can edit loan amount, interest, installments, status, and notes at any time.

6. Customer Loan Tracking

Customers can view current status, loan amount, interest rate, installments, payment schedule (if added), and notes in read-only mode.

7. Chat & Communication

Real-time encrypted messaging between customer and employee with file sharing (images, PDFs, documents). Admin can monitor all conversations. Internal employee chat rooms (e.g., Contracts, Follow Up, Receipts) are managed by Admin.

8. Broadcast Messaging

Admin can send offers, announcements, or notices to all employees, all customers, or selected groups via system notifications.

9. Dashboards

Admin Dashboard: analytics, user management, loan management, chat monitoring, announcements, language control.
Employee Dashboard: assigned customers, loan updates, status management, chat, internal rooms, notifications.
Customer Dashboard: loan details (read-only), loan status, chat, file upload, notifications.

10. Workflows

Customer Assignment: Admin creates customer → assigns employee → both receive notification.
Loan Update: Employee edits loan → status updated → customer notified → admin can review.
Chat: Customer/Employee message → real-time notification → stored securely.

Broadcast: Admin creates announcement → selects audience → notification delivered.

Final Summary

This system provides a structured digital platform for reflecting loan data, managing communications, and maintaining status updates without performing automatic calculations. It ensures administrative control, operational clarity, and secure interaction between employees and customers.
If you want, I can now convert this into a 1-page executive summary for proposal submission.

Technologies:

We will be using Next JS with Node JS based backend mysql for the database and for push notifications we will use firebase and its services.

3rd Party Apis:

Any third-party API used google translation api, or services like Firebase, Servers, Databases and will be provided by the client. APIs cost will be paid by the client; its charges will not be included in the development cost discussed.

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Costing

Development Phases	Price($)
Scope Document, Web app frontend complete.	500$
Backend Integration	500$
Chat, Push notifications and Deployment	500$
Total	**1500$**

Note:

Price Quoted above is for the scope of the project discussed in the document. We can add additional features if required, for that you will be charged additional budget to the proposed above. Both hosting, domain and database required accounts. Its cost will not be included in the mentioned app budget.

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Development Stages

Stage	Description
Information Gathering	We will make a thorough understanding of all the app features by having a formal meeting with you via call or on text communication.
UI/UX Design	Design of all the user Application Screens in accordance with finalized Color scheme and functionality
IOS Prototyping	Front End Presentation of the finalized design in IOS Device coded in Dart or Swift UI(Cupertino Design)
Android Prototyping	Front End Presentation of the finalized design in Android Device coded in Dart / Java with xml (Material Design)
Database Design	Making Database Structure; all the tables, hierarchy and defining all the roles
Fetching Data response from Admin Panel to the Application	Service calling from admin panel and sending response in JSON to the application backend
Integration of Admin Panel with apps	Implementation of services and making Admin Panel and Apps integrated
QA Testing	Detailed Testing and Debugging of all the App, and Admin Panel Features
Deployment	Deploying Data to your server and Apps to your Play store and App Store Accounts

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System & App Architecture Design

System & App Architecture Design is often the most critical phase of any software development effort. In many cases, the quality of the finished software is directly proportional to the quality of the initial design. A particularly appropriate metaphor is... Ask yourself this question: What quality of house would you get if you built it with no blueprints?
Custom software development is an investment. As with any substantial investment, planning is crucial. However, there is a trade-off. It is possible to over-design. This leads to lost time and money. We have the experience to strike the proper balance.

Technical Support Packages

Technical Support includes

- Security Updates: Security is the foremost priority of all the online systems in today's World. Our technical team will make sure that all your security certificates are up to date and if any update is needed in the system, we will be doing it right away to make sure no malicious activity is harming the system.
- Payment Gateway Updates: For receiving payments, we will be using Software Development Kits (SDKs) which will be responsible to link your platform with the banks and manage all the payments. To make sure the payment process is smooth, all the SDKs need to be updated. Our support team will be looking for updates on a weekly basis, and we will be providing you with updates for all the payment gateway SDKs to keep the transaction process smooth.
- Bug Fixing: The platform we will be creating will be a combination of lots of codes. If any part of the code malfunctions, the system can get crashed or any part of it can malfunction, if any bug is reported by any user, our team will be providing you support and assistance on the bug fixing in accordance to the package you choose from the packages mentioned below.

Support Packages	Price($)
PREMIUM APP SUPPORT WITHIN 24-48 HOURS	$350/month
YEARLY PACKAGE	3500$/year

Support & Maintenance

- After project delivery we will provide you free app support of up to 30 days.
- The platform we will be creating will contain a bulk of codes. If any part of the code malfunctions the system can get crashed. If a bug is reported by any of your clients, our team will be providing you assistance with bug fixing depending upon the package you will choose for support in the future.
- Updates Will be charged separately.

Servers And Databases

Services of firebase will be used along with its cloud firestore and cloud functions for backend development. Once development starts we will send you a list of firebase pricing plans. So that you can choose a plan which suits you.

Add ons

Servers

For storing the admin panel, website and data of the apps, we need to have a server where everything will be hosted. As the user base will not be very high in start, we recommend using Digital Ocean hosting and choosing linux based droplets. Once your app and business grows we can switch it's packages according to our requirements.

The server will be registered on your/your company's name and you'll be bearing the payment and we can buy it and set it up for you. Following are the cost details of the servers we can use;

DigitalOcean (dedicated): https://www.digitalocean.com/pricing/

(Note: Will use the introductory compute droplet ($5/month) in start then scale as website traffic increases)

AWS (dedicated): https://aws.amazon.com/ec2/pricing/

(Note: Will use the Free tier for development and starting and then scale up as website traffic increases)

Bluehost (shared): https://www.bluehost.com/hosting/shared#pricing-cards

Domain

To host the website and backend of your app, we will need to have a domain name. If you already have a domain name, we can use that otherwise we can buy a domain for you.

Future Releases

IT is one of the fastest-growing industries nowadays. As the trends change, its platforms like web, app development platforms also change. We are here to take care of your application and make sure it's up to date. If you want any updates or the addition of new modules we are always here to update your application.

The price of the release will also be discussed when requirements will be discussed for future releases.

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